1.7.1 ▶ ▶ ▶ Hydrostatic Pressure

For ideal fluids, there is a direct correlation between pressure and depth within the fluid. To see this relation, consider a container of fluid with density ρ in hydrostatic equilibrium in an environment with ambient pressure P_0. Since no parts of the fluid are moving (at a macroscopic scale), all parts of the fluid must be in mechanical equilibrium. From this container, consider the cylinder of fluid with cross-sectional area A and height h outlined in the figure.

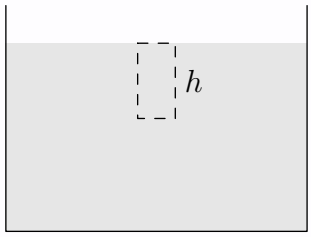

Fig. 1.17

There are three forces acting on this cylinder: an upward force from the hydrostatic pressure at depth h, a downward force from the ambient pressure in the environment, a downward force due to the weight of the fluid. Since the net force on the cylinder is zero, we have the following relation:

$$P(h)A = P_0A + mg.$$

Where $P(h)$ is the pressure of the fluid at depth h. Rewriting the mass of the cylinder using the density of the fluid yields

$$P(h)A = P_0A + \rho Ah \cdot g$$

$$\boxed{P(h) = P_0 + \rho gh} \tag{1.108}$$

Notice that this equation puts no constraint on the value or direction of g. So, if a fluid is placed in an accelerated reference frame where the g vector is no longer parallel, the pressure gradient will be anti-parallel to the direction of the g vector, and not necessarily vertical[18]. ★

[18] An argument can be made here that "vertical" is defined as the direction opposite to the g vector, but at the end of the day its quite semantical. Here we just consider a reference frame external to the fluid, where "vertical" isn't parallel to the effective gravity in the fluid.